(ii) **Linear polynomial** A polynomial of degree 1, is called linear polynomial. e.g. $9x+7$, $x-9$, $x+2$ etc.

(iii) **Quadratic polynomial** A polynomial of degree 2, is called quadratic polynomial.

The general form of quadratic polynomial in x is ax^2+bx+c, where a, b and c are real and $a \neq 0$.

e.g. $x^2-7x+12, 5y^2-7, z^2-4$ etc., are all quadratic polynomials.

(iv) **Cubic polynomial** A polynomial of degree 3, is called cubic polynomial.

The general form of a cubic polynomial in x is ax^3+bx^2+cx+d, where a, b, c and d are real and $a \neq 0$.

e.g. $3x^3+2x^2+7x-1$, $8y^3+5y+2$, x^3-8 etc., are all cubic polynomials.

(v) **Biquadratic polynomial** A polynomial of degree 4, is called a biquadratic polynomial.

The general form of a biquadratic polynomial in x is $ax^4+bx^3+cx^2+dx+e$, where a, b, c, d and e are real numbers and $a \neq 0$.

e.g. $6x^4+3x^3+2x^2+x+2$, x^4-8 etc.,

are all biquadratic polynomials.

Based on Number of Terms

(i) **Monomial** A polynomial containing only one term is called a monomial. e.g. $6x^2$

(ii) **Binomial** A polynomial containing two terms is called a binomial. e.g. $5x^3+7xy$

(iii) **Trinomial** A polynomial containing three terms is called a trinomial. e.g. $7x^2y+5x+6$

FACTORISATION

To express a polynomial as the product of other polynomials of degree less than that of the given polynomial is called as factorisation.

e.g. $x^2-49 = x^2-7^2 = (x-7)(x+7)$

Important Identities

- $(a^2-b^2) = (a+b)(a-b)$
- $(a+b)^2 = a^2+b^2+2ab$ and $(a-b)^2 = a^2+b^2-2ab$
- $(a+b)^2-(a-b)^2 = 4ab$
- $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$
- $(a+b)^3 = a^3+b^3+3ab(a+b)$
- $(a-b)^3 = a^3-b^3-3ab(a-b)$
- $(a^3+b^3) = (a+b)(a^2+b^2-ab)$
- $(a^3-b^3) = (a-b)(a^2+b^2+ab)$
- $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$
- $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-ac-bc)$
- If $a+b+c=0$, then $a^3+b^3+c^3 = 3abc$
- $a^4+a^2b^2+b^4 = (a^2+ab+b^2)(a^2-ab+b^2)$

Methods of Factorisation

These methods are given below

Factorisation by Taking Out the Common Factor

If each term of an expression has a common factor, take out the common factors.

EXAMPLE 1. Factorise value of $4(3a-2b)^2-5(3a-2b)$ is

a. $(3a-2b)(12a-8b-5)$ b. $(2a-3b)(12a-8b)$

c. $(2a-3b)(8a-12b-5)$ d. None of these

Sol. a. $4(3a-2b)^2-5(3a-2b) = (3a-2b)[4(3a-2b)-5]$

$= (3a-2b)[12a-8b-5]$

Factorisation by Grouping

Sometimes in a given polynomial, it is not possible to take out a common factor directly. However, on rearranging the terms of the polynomial and grouping them such that all the terms have a common factor, the polynomial can be easily factorised.

EXAMPLE 2. Factorise $x^2+y-xy-x$

a. $(x-y)(x+1)$ b. $(x-y)(x^2-1)$

c. $(x+y)(x^2+1)$ d. $(x-y)(x-1)$

Sol. d. **Method I** $x^2+y-xy-x = x^2-x+y-xy$

$= x(x-1)+y(1-x) = x(x-1)-y(x-1) = (x-1)(x-y)$

Method II $x^2+y-xy-x = x^2-xy+y-x$

$= x(x-y)-1(x-y) = (x-y)(x-1)$

EXAMPLE 3. Factorise $x^2+\frac{4}{x^2}+4-2x-\frac{4}{x}$

a. $\left(x-\frac{2}{x}\right)\left(x+\frac{2}{x}+2\right)$ b. $\left(x-\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$

c. $\left(x+\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$ d. None of these

Sol. c. $\left(x^2+\frac{4}{x^2}+4-2x-\frac{4}{x}\right) = \left(x^2+\frac{4}{x^2}+4\right)-2\left(x+\frac{2}{x}\right)$

$= \left(x+\frac{2}{x}\right)^2-2\left(x+\frac{2}{x}\right) = \left(x+\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$

Factorisation of Perfect Square Polynomials

If the polynomial is given as the perfect square quadratic polynomial, then use the following Identities to factorise it.

- $a^2+2ab+b^2=(a+b)^2$
- $a^2-2ab+b^2=(a-b)^2$

EXAMPLE 4. Factorise $16x^2-8x+1$

a. $(2x-1)^2$ b. $(x-4)^2$ c. $(x-2)^2$ d. $(4x-1)^2$

Sol. d. $16x^2-8x+1=(4x)^2-2(4x)+1^2=(4x-1)^2$
$=(4x-1)(4x-1)$

EXAMPLE 5. Factorise $x^2-2\sqrt{5}x+5$

a. $(x\sqrt{5}-1)^2$ b. $(x-5)^2$ c. $(x+\sqrt{5})(x-\sqrt{5})$ d. $(x-\sqrt{5})^2$

Sol. d. $x^2-2\sqrt{5}x+5=x^2-2\sqrt{5}x+(\sqrt{5})^2=(x-\sqrt{5})^2$
$=(x-\sqrt{5})(x-\sqrt{5})$

Factorising the Difference of Two Squares

In case, the polynomial is given in the form of a^2-b^2, to evaluate it, following identity is applied.

- $a^2-b^2=(a-b)(a+b)$

EXAMPLE 6. Factorise $2a^5-32a$

a. $(a^2+4)(a-2)(a+2)$ b. $2a(a^2+4)(a-2)(a+2)$
c. $2a(a^2+4)$ d. None of these

Sol. b. $2a^5-32a=2a(a^4-16)$
$=2a[(a^2)^2-(4)^2]=2a(a^2+4)(a^2-4)$
$=2a(a^2+4)(a^2-2^2)=2a(a^2+4)(a-2)(a+2)$

Factorisation of Quadratic Polynomials

Quadratic polynomials of the type ax^2+bx+c, where $a\neq 0$, a and b are coefficients of x^2 and x respectively and c is constant, can be factorised by splitting the middle term. We find two numbers p and q such that

$p+q=b$ and $pq=ac$, then

$$ax^2+bx+c=ax^2+(p+q)x+c=ax^2+px+qx+c$$

EXAMPLE 7. Factorise $x^2+9x+14$

a. $(x+2)(x+7)$ b. $(x-2)(x-7)$
c. $(x+2)(x-7)$ d. $(x-2)(x+7)$

Sol. a. $x^2+9x+14=x^2+(7+2)x+14=x^2+7x+2x+14$
$=x(x+7)+2(x+7)=(x+7)(x+2)$

EXAMPLE 8. Factorise $4\sqrt{3}x^2+5x-2\sqrt{3}$

a. $(2x+\sqrt{3})(4x-\sqrt{3})$ b. $(2x-\sqrt{3})(4x+\sqrt{3})$
c. $(\sqrt{3}x+2)(4x-\sqrt{3})$ d. None of these

Sol. c. $4\sqrt{3}x^2+5x-2\sqrt{3}=4\sqrt{3}x^2+8x-3x-2\sqrt{3}$
$[\because 4\sqrt{3}\times(-2\sqrt{3})=-24=8\times(-3)]$
$=4x(\sqrt{3}x+2)-\sqrt{3}(\sqrt{3}x+2)$
$=(\sqrt{3}x+2)(4x-\sqrt{3})$

Factorisation of Sum and Difference of Cubes

In case, the polynomial is given in the form of a^3+b^3 or a^3-b^3. To factorise it, following identities can be applied.

- $a^3+b^3=(a+b)(a^2-ab+b^2)$
- $a^3-b^3=(a-b)(a^2+ab+b^2)$

EXAMPLE 9. Factorise $(2x+3y)^3-(2x-3y)^3$

a. $18(4x^2+3y^2)$ b. $18y(3x^2+4y^2)$
c. $xy(4x^2+3y^2)$ d. $18y(4x^2+3y^2)$

Sol. d. $(2x+3y)^3-(2x-3y)^3$

Put $2x+3y=a$ and $2x-3y=b$

$\Rightarrow$ $a^3-b^3=(a-b)(a^2+ab+b^2)=(2x+3y-2x+3y)$
$[(2x+3y)^2+(2x+3y)(2x-3y)+(2x-3y)^2]$
$=6y[4x^2+9y^2+12xy+4x^2-9y^2+4x^2+9y^2-12xy]$
$=6y[12x^2+9y^2]=18y[4x^2+3y^2]$

Hence, $(2x+3y)^3-(2x-3y)^3=18y[4x^2+3y^2]$

Factorisation of Polynomial of form $a^3+b^3+c^3-3abc$

Here, it is easy to use.

- $a^3+b^3+c^3-3abc$
$=(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$
- If $a+b+c=0$, then $a^3+b^3+c^3=3abc$

EXAMPLE 10. Factorise $x^3+27y^3+8x^3-18xyz$

a. $(x+3y+2z)(x^2+9y^2+4z^2+3xy+6yz+2xz)$
b. $(x+3y+2z)(x^2+9y^2+4z^2-3xy-6yz-2xz)$
c. $(x-3y-2z)(x^2+9y^2-4z^2-3xy+6yz+2xz)$
d. None of the above

Sol. b. $x^3+27y^3+8z^3-18xyz$
$=x^3+(3y)^3+(2z)^3-3(x)(3y)(2z)$
$=(x+3y+2z)[x^2+9y^2+4z^2-x(3y)-3y(2z)-x(2z)]$
$=(x+3y+2z)([x^2+9y^2+4z^2-3xy-6yz-2xz])$

EXAMPLE 11. Factorise
$(2x-3y)^3+(3y-5z)^3+(5z-2x)^3$

a. $3(2x-3y)(3y-5z)(5z-2x)$
b. $(2x-3y)(3y-5z)(5z-2x)$
c. $3(2x-3y)(3y+5z)(5z-2x)$
d. None of the above

Sol. **a.** Here, $2x-3y=a$, $3y-5z=b$ and $5z-2x=c$

$\because \quad a+b+c=0 \Rightarrow a^3+b^3+c^3=3abc$

$\therefore \quad (2x-3y)^3+(3y-5z)^3+(5z-2x)^3$

$$= 3(2x-3y)(3y-5z)(5z-2x)$$

Remainder Theorem

Let $p(x)$ be a polynomial of degree n greater than or equal to 1 (i.e. $n \geq 1$) and a be any real number. If $p(x)$ is divided by the linear polynomial $(x-a)$, then the remainder is $p(a)$. Remainder can be evaluated by substituting $x-a=0$, i.e. $x=a$ in $p(x)$.

EXAMPLE 12. The remainder when $12x^3-13x^2-5x+9$ is divided by $(3x+2)$ is

a. 1 **b.** 2 **c.** 3 **d.** 0

Sol. **c.** Let $p(x)=12x^3-13x^2-5x+9$ and $q(x)=3x+2$

When $p(x)$ is divided by $(3x+2)$, then remainder is $p\left(\frac{-2}{3}\right)$.

Now, $p\left(\frac{-2}{3}\right)=12\left(\frac{-2}{3}\right)^3-13\left(\frac{-2}{3}\right)^2-5\left(\frac{-2}{3}\right)+9$

$$=12\times\left(\frac{-8}{27}\right)-13\times\frac{4}{9}+\frac{10}{3}+9=3$$

Hence, the required remainder is 3.

Factor Theorem

Let $p(x)$ be a polynomial, if degree $n \geq 1$ and a be any real number. Then,

(i) If $p(a)=0$, then $(x-a)$ is a factor of $p(x)$.

(ii) If $(x-a)$ is a factor of $p(x)$, then $p(a)=0$.

EXAMPLE 13. For what values of k will $4x^5+9x^4-7x^3-5x^2-4kx+3k^2$ contain $x-1$ as a factor?

a. $3, -\frac{1}{2}$ **b.** $3, -1$ **c.** $0, \frac{1}{3}$ **d.** $1, \frac{1}{3}$

Sol. **d.** Given, $P(x)=4x^5+9x^4-7x^3-5x^2-4kx+3k^2$

Since, $(x-1)$ is a factor of $P(x)$. $\therefore \quad P(1)=0$

$\Rightarrow 4\times(1)^5+9\times(1)^4-7\times(1)^3-5\times(1)^2-4\times k\times(1)+3\times k^2=0$

$\Rightarrow \quad 3k^2-4k+1=0 \Rightarrow 3k^2-3k-k+1=0$

$\Rightarrow \quad (3k-1)(k-1)=0 \Rightarrow k=\frac{1}{3}, 1$

Division Synthetic Algorithm

If $p(x)$ and $g(x)$ any two polynomials with $g(x)\neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that

$$p(x)=g(x)\times q(x)+r(x)$$

i.e. Dividend = (Divisor × Quotient) + Remainder

where, $r(x)=0$ or degree of $r(x)<$ degree of $g(x)$, then we say that $p(x)$ divided by $g(x)$, gives $q(x)$ as quotient and $r(x)$ as remainder.

If the remainder $r(x)$ is zero, we say that divisor $g(x)$ is a factor of $p(x)$.

EXAMPLE 14. If two factors of $a^4-2a^3-9a^2+2a+8$ are $(a+1)$ and $(a-1)$, then what are the other two factors?

a. $(a-2)$ and $(a+4)$ **b.** $(a+2)$ and $(a+4)$

c. $(a+2)$ and $(a-4)$ **d.** $(a-2)$ and $(a-4)$

Sol. **c.** Let $f(a)=a^4-2a^3-9a^2+2a+8$

and $(a+1)$ and $(a-1)$ are two factors of $f(a)$.

Now, by division method,

$$\begin{array}{l}
a^2-1)\; a^4-2a^3-9a^2+2a+8\; (a^2-2a-8 \\
\quad\quad\;\; a^4 \quad\quad\;\; - a^2 \\
\hline
\quad\quad\;\; -2a^3-8a^2+2a+8 \\
\quad\quad\;\; -2a^3 \quad\quad\;\; 2a \\
\hline
\quad\quad\quad\quad -8a^2 \quad +8 \\
\quad\quad\quad\quad -8a^2 \quad +8 \\
\hline
\quad\quad\quad\quad\quad \times
\end{array}$$

So, the required factor of $f(a)$ is (a^2-2a-8).

Further factorise it by splitting the middle term, i.e.

$$a^2-2a-8=a^2-4a+2a-8$$
$$=a(a-4)+2(a-4)=(a+2)(a-4)$$

Hence, $(a+2)$ and $(a-4)$ are other two factors of $f(a)$.

Factorisation of Polynomials Using Factor Theorem

To factorise the polynomial, follow the following steps

(i) Let $p(x)$ be a given polynomial.

(ii) find all the possible factors of constant term in $p(x)$.

(iii) Take any one of the factors, say (α) and find $p(\alpha)$. If $p(\alpha)=0$ the $(x-\alpha)$ is a factor of $p(x)$.

(iv) Divide the polynomial $p(x)$ by $(x-\alpha)$ to find its all other factors.

EXAMPLE 15. Factorise $2x^3-5x^2-19x+42$.

a. $(x-2)(2x-7)(x+3)$ **b.** $(x+2)(2x+7)(x-3)$

c. $(x-1)(3x-7)(x-4)$ **d.** $(2x+3)(4x-7)(x-1)$

Sol. **a.** Let $p(x)=2x^3-5x^2-19x+42$

Hence, constant term is 42 and its all factors are $=\pm 1$, ± 2, ± 3, ± 6, ± 7, ± 14, ± 21 and ± 42.

At $x=1$, $\quad p(1)=2(1)^3-5(1)^2-19+42$

$$=2-5-19+42=20\neq 0$$

So, $(x-1)$ is not a factor of $p(x)$.

At $x=2$, $p(2)=2(2)^3-5(2)^2-19(2)+42=58-58=0$

So, $(x-2)$ is a factor of $p(x)$.

Now, on dividing $p(x)$ by $(x-2)$, we get $(2x^2-x-2)$ as quotient i.e. other factor of $p(x)$.

So, $\quad p(x)=(x-2)[2x^2-x-21]$

$$=(x-2)(2x^2-7x+6x-21)$$
$$=(x-2)[x(2x-7)+3(2x-7)]$$

$\therefore \quad p(x)=(x-2)(2x-7)(x+3)$

PRACTICE EXERCISE

1. If $x = 5, y = 3$ and $z = 2$, then the value of $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$ is

(a) 125 (b) 0 (c) –25 (d) 10

2. The factors of $x^2 - 2\sqrt{3}x + 3$ are

(a) $(x + \sqrt{3})^2$ (b) $(x - \sqrt{3})^2$

(c) $(x + \sqrt{3})(x - \sqrt{3})$ (d) $(x + 2)(x + \sqrt{3})$

3. The factors of $(a^4b^4 - 16c^4)$ are

(a) $4(a^2b^2 + c^2)(ab - 2c)(ab + 2c)$

(b) $(a^2b^2 - 4c^2)(ab + 2c)^2$

(c) $(a^2b^2 + 4c^2)(ab + 2c)(ab - 2c)$

(d) $(a^2b^2 - 4c^2)^2(ab + 2c)(ab + 4c)$

4. The factors of $(x^8 - y^8)$ are

(a) $(x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$

(b) $(x^2 + y^2)^2(x + y)(x - y)$

(c) $(x^4 + y^4)(x^2 + y^2)^2$

(d) $(x^2 + y^2)(x - y)^2$

5. The factors of $\left(a^2 + a + \frac{1}{4}\right)$ are

(a) $\left(a + \frac{1}{2}\right)^2$ (b) $\left(a + \frac{1}{2}\right)(a + 2)$

(c) $\left(a + \frac{1}{2}\right)\left(a - \frac{1}{2}\right)$ (d) $\left(a - \frac{1}{2}\right)^2$

6. The factors of $8 - 4x - 2x^3 + x^4$ are

(a) $(2 - x)(4 - x^3)$ (b) $(2 + x)(4 - x^3)$

(c) $(2 + x)(3 - x^3)$ (d) $(2 - x)(x^3 - 4)$

7. The factors of $(a^2 - b^2 - 4ac + 4c^2)$ are

(a) $(a + 2c + b)(a - 2c - b)$ (b) $(a - 2c + b)(a - 2c - b)$

(c) $(a - 2b + c)(a + b + 2c)$ (d) $(a - 2b)(a + 2b + 2c)$

8. What are the factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$?

(a) $(x - 2y - 4)$ and $(x - 2y + 2)$

(b) $(x - y + 2)$ and $(x - 4y + 4)$

(c) $(x - y + 2)$ and $(x - 4y - 4)$

(d) $(x + 2y - 4)$ and $(x + 2y + 2)$

9. Factorise $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$

(a) $\left(a + \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2}\right) - 2\left(a - \frac{1}{a}\right)$

(b) $\left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) + 2\left(a - \frac{1}{a}\right)$

(c) $\left(a + \frac{1}{a}\right)\left(a - \frac{1}{a}\right)$

(d) None of the above

10. What are the factors of $\left(\frac{1}{3}x^2 - 2x - 9\right)$?

(a) $\frac{1}{3}(x - 9)(x + 3)$ (b) $\frac{1}{3}(x - 9)(x - 3)$

(c) $\frac{1}{3}(x + 9)(x + 3)$ (d) $\frac{1}{3}(x + 9)(x - 3)$

11. What are the factors of $(6\sqrt{3}x^2 - 47x + 5\sqrt{3})$?

(a) $(3\sqrt{3}x + 5\sqrt{3})(2x - 5\sqrt{3})$

(b) $(3\sqrt{3}x - 5\sqrt{3})(x\sqrt{3}x + 1)$

(c) $(2x + 5\sqrt{3})(3\sqrt{3}x + 1)$

(d) $(2x - 5\sqrt{3})(3\sqrt{3}x - 1)$

12. The factors of $(8a^3 + 125b^3 - 64c^3 + 120abc)$ are

(a) $(2a + 5b - 4c)(2a + 5b + 4c)$

(b) $(2a - 5b - 4c)(2a + 5b + 4c)$

(c) $(2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$

(d) $(2a + 5b + 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$

13. If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then

(a) $x + y + z = 0$ (b) $(x + y + z)^3 = 27xyz$

(c) $x + y + z = 3xyz$ (d) $x^3 + y^3 + z^3 = 0$

14. If $x + \frac{1}{x} = \sqrt{5}$, then the value of $x^3 + \frac{1}{x^3}$ is

(a) $8\sqrt{5}$ (b) $2\sqrt{5}$ (c) $5\sqrt{5}$ (d) $7\sqrt{5}$

15. If $\left(x + \frac{1}{x}\right) = 6$, then $\left(x^2 + \frac{1}{x^2}\right)$ is equal to

(a) 32 (b) 38 (c) 34 (d) 44

16. If $x - \frac{1}{x} = \frac{1}{3}$, then what is $9x^2 + \frac{9}{x^2}$ equal to?

(a) 18 (b) 19 (c) 20 (d) 21

17. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 26$, then what is $ab + bc + ca$ equal to?

(a) 0 (b) 2 (c) 4 (d) 5

18. If $(x^4 + x^{-4}) = 322$, then what is one of the value of $(x - x^{-1})$?

(a) 18 (b) 16 (c) 8 (d) 4

19. If $a + b + c = 0$, then what is the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

(a) – 3 (b) 0 (c) 1 (d) 3

20. If $a + b + c = 10$ and $ab + bc + ca = 31$, then the value of $a^2 + b^2 + c^2$ is

(a) 48 (b) 38 (c) 28 (d) 18

21. The factors of $x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x}$ are

(a) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} + 2\right)$ (b) $\left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)$

(c) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$ (d) $\left(x + \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$

22. The factors of $(a^2 - b^2)(c^2 - d^2) - 4abcd$ are

(a) $(ac + bd + bc + ad)(ac - bd - bc - ad)$
(b) $(ac - bd - bc + ad)(ac + bd + dc + ab)$
(c) $(ac - bd + bc + ad)(ac - bd - bc - ad)$
(d) $(ac - bd - bc - ad)(ac + bd + dc + ab)$

23. If 'a' is an integer such that $a + \frac{1}{a} = \frac{17}{4}$, then the value of $\left(a - \frac{1}{a}\right)$ is

(a) 4 (b) $\frac{13}{4}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$

24. $x^4 + 4y^4$ is divisible by which one of the following?

(a) $(x^2 + 2xy + 2y^2)$ (b) $(x^2 + 2y^2)$
(c) $(x^2 - 2y^2)$ (d) None of these

25. If $x(x + y + z) = 9$, $y(x + y + z) = 16$ and $z(x + y + z) = 144$, then x equal to

(a) $\frac{9}{5}$ (b) $\frac{9}{7}$ (c) $\frac{9}{13}$ (d) $\frac{16}{13}$

26. If $x = (b - c)(a - d)$, $y = (c - a)(b - d)$ and $z = (a - b)(c - d)$, then what is $x^3 + y^3 + z^3$ equal to?

(a) xyz (b) $2xyz$ (c) $3xyz$ (d) $-3xyz$

27. If the expression $(px^3 + x^2 - 2x - q)$ is divisible by $(x - 1)$ and $(x + 1)$, what are the values of p and q respectively?

(a) 2, – 1 (b) – 2, 1 (c) – 2, – 1 (d) 2, 1

28. If the expression $px^3 + 3x^2 - 3$ and $2x^3 - 5x + p$ when divided by $x - 4$ leave the same remainder, then what is the value of p?

(a) – 1 (b) 1 (c) – 2 (d) 2

29. Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n$, where $a_0, a_1, a_2 \ldots a_n$ are constants. If $f(x)$ is divided by $ax - b$, the remainder is

(a) $f\left(\frac{b}{a}\right)$ (b) $f\left(\frac{-b}{a}\right)$ (c) $f\left(\frac{a}{b}\right)$ (d) $f\left(\frac{-a}{b}\right)$

30. Which one of the following statements is correct?

(a) Remainder theorem is a special case of factor theorem
(b) Factor theorem is a special case of remainder theorem
(c) Factor theorem and remainder theorem are two independent results
(d) None of the above

31. What is $x(y - z)(y + z) + y(z - x)(z + x) + z(x - y)(x + y)$ equal to?

(a) $(x + y)(y + z)(z + x)$ (b) $(x - y)(x - z)(z - y)$
(c) $(x + y)(z - y)(x - z)$ (d) $(y - x)(z - y)(x - z)$

32. If the remainder of the polynomial $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ when divided by $(x - 1)$ is 1, then which one of the following is correct?

(a) $a_0 + a_2 + \cdots = a_1 + a_3 + \cdots$
(b) $a_0 + a_2 + \cdots = 1 + a_1 + a_3 + \cdots$
(c) $1 + a_0 + a_2 + \cdots = -(a_1 + a_3 + \cdots)$
(d) $1 - a_0 - a_2 - \cdots = a_1 + a_3 + \cdots$

33. If u, v and w are real numbers such that $u^3 - 8v^3 - 27w^3 = 18uvw$, then which one of the following is correct?

(a) $u - v + w = 0$ (b) $u = -v = -w$
(c) $u - 2v = 3w$ (d) $u + 2v = -3w$

34. The value of $(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8})(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})$ is

(a) $(a + a^{-1})$ (b) $(a - a^{-1})$ (c) $(a^2 + a^{-2})$ (d) $(a^2 - a^{-2})$

35. The polynomial $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$, when divided by $(x - 1)$ and $(x + 1)$ leaves the remainders 5 and 19, respectively. The values of a and b are

(a) $a = 4, b = 3$ (b) $a = 3, b = 4$
(c) $a = 5, b = 8$ (d) $a = 9, b = 7$

36. If the expression $ax^2 + bx + c$ is equal to 4 when $x = 0$, leaves a remainder 4 when divided by $x + 1$ and a remainder 6 when divided by $x + 2$., then the values of a, b and c are respectively:

(a) 1, 1, 4 (b) 2, 2, 4 (c) 3, 3, 4 (d) 4, 4, 4

37. If $(x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2})$ is divided by $x^{1/2} - y^{1/2}$ the quotient is

(a) $x - y$ (b) $x + y$ (c) $x^{1/2} + y^{1/2}$ (d) $x^2 - y^2$

38. If, $9x^2 + 3px + 6q$ when divided by $3x + 1$ leaves a remainder $-\frac{3}{4}$ and $qx^2 + 4px + 7$ is exactly divisible by $x + 1$, then the values of p and q respectively are

(a) 0, 7 / 4 (b) –7 / 4, 0
(c) $\frac{7}{4}$, 0 (d) None of these

39. If $a + b + c = 1$, then which of the following statements are true?

I. $(a + b)(b + c)(c + a) = bc + ac + ab - abc$
II. $a^2 + b^2 - c^2 + 2ab = a + b - c$

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

40. If $x + \frac{1}{x} = 4$, then the value of expression

I. $x^3 + \frac{1}{x^3} = 52$ II. $x = 2 + \sqrt{3}$

Which of the following is correct?

(a) Only I (b) Only II
(c) Neither I nor II (d) Both I and II